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ECONOMIC TRANSITION AND GROWTH

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SUMMARY

Some extensions of neoclassical growth models are discussed that allow for cross-section heterogeneity among economies and evolution in rates of technological progress over time. The models offer a spectrum of transitional behavior among economies that includes convergence to a common steady-state path as well as various forms of transitional divergence and convergence. Mechanisms for modeling such transitions, measuring them econometrically, assessing group behavior and selecting subgroups are developed in the paper. Some econometric issues with the commonly used augmented Solow regressions are pointed out, including problems of endogeneity and omitted variable bias which arise under conditions of transitional heterogeneity. Alternative regression methods for analyzing economic transition are given which lead to a new test of the convergence hypothesis and a new procedure for detecting club convergence clusters. Transition curves for individual economies and subgroups of economies are estimated in a series of empirical applications of the methods to regional US data, OECD data and Penn World Table data. Copyright © 2009 John Wiley & Sons, Ltd.

> The legacy of economic growth that we have inherited from the industrial revolution is an irreversible gain to humanity, of a magnitude that is still unknown. ... The legacy of inequality, the concomitant of this gain, is a historical transient. Lucas, (2002, pp. 174–175)

1. INTRODUCTION

In his study of the growth of nations in the world economy over the last 250 years, Lucas (2002) argues that the enormous income inequality that followed in the swath of the industrial revolution has now peaked. Instead, in the 21st century, as countries increasingly participate in the economic benefits of industrialization, this income inequality will prove to be a historical transient. Building on a model of Becker *et al*. (1990), Lucas develops a theory that seeks to explain the transition that has occurred in the world economy from the stagnant steady-state economies that persisted until around 1800 to modern economies that experience sustained income growth. Human capital accumulation is posited as the engine of this growth, and the mechanism by which it is accomplished comes by way of a demographic transition that emerges from the inclusion of fertility decision making into the theory of growth. These arguments involve two forms of transition: a primary economic transition involving the move toward sustained economic growth, and a secondary, facilitating demographic transition associated with declining fertility.

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Lucas supports the arguments by some descriptive data analysis that document the transitions and suggest the emergent transience in income inequality mentioned in the headnote quotation.

The present paper looks at the phenomenon of 'economic transition' from an econometric perspective. We ask two main questions and then proceed to develop an econometric methodology for studying issues of economic transition empirically. The first question concerns neoclassical economic growth and asks if the model has the capacity to generate transitional heterogeneity of economic growth patterns across countries that are consistent with historical income inequality while still allowing for some form of ultimate growth convergence. Such behavior would have to accommodate transient divergence in growth patterns. So, a subsidiary question relates to the conditions under which such transitional economic divergence could occur and how it might be parameterized and evaluated empirically.

In seeking to address the first question, we use a neoclassical growth model under heterogeneous technological progress, as suggested by Barro and Sala-i-Martin (1997), Howitt and Mayer-Foulkes (2005) and Parente and Prescott (1994). Within this framework we develop a nonlinear dynamic factor model for log per capita real income of the form

$$\log y_{it} = a_{it} + x_{it}t = b_{it}\mu_t \tag{1}$$

where the component a_{it} embodies transitional dynamics for real effective capital and the component $x_{it}t$ captures the idiosyncractic time paths of technological progress. Both components are permitted to be heterogeneous across individuals and over time. The dynamic factor formulation $b_{it}\mu_t$ involves a growth component, μ_t, that is common across individuals (for instance, μ_t may represent a proxy for commonly available world technology), and individual transition factors (b_{it}) that measure how individual economic performance relates over time to μ_t.

In contrast to (1), transitional dynamics for log per capita real income are often expressed in the form

$$\log y_{it} = \log \tilde{y}_i^* + \log A_0 + [\log \tilde{y}_{i0} - \log \tilde{y}_i^*]e^{-\beta t} + xt \tag{2}$$

where $\log \tilde{y}_{i0}^*$ and $\log \tilde{y}_i^*$ denote initial and steady-state levels of effective log per capita real income, and $\log A_0$ is the initial log technology. In this model, where the transition parameter β and the growth rate x are homogeneous, neoclassical theory does not naturally accommodate such enormous differences in observed income growth as the world economy has witnessed in the success of the Asian Dragons or the growth disasters of sub-Saharan Africa in relation to other developing countries. However, when we permit cross-sectional and temporal heterogeneity in these parameters—replacing β and x in (2) by β_{it} and x_{it}—neoclassical growth can provide for such forms of transitional cross-sectional divergence. With these extensions, the model may also allow for ultimate growth convergence, thereby making cross-country income inequality a transient phenomena, as argued by Lucas.

In such an extended model, the speed of convergence parameter β_{it} may reasonably be regarded as an increasing function of technological progress x_{it}. Accordingly, poor economies with a low level of technological accumulation may begin with a low β_{it} and a correspondingly slow speed of convergence. As such countries learn faster (e.g., from improvements in education and the diffusion of technology), their x_{it} rises and may exceed the rate of technological creation in rich nations. So, β_{it} rises and the speed of convergence of these economies begins to accelerate. Conversely, if a poor country responds slowly to the diffusion of technology by learning slowly or through suffering a major economic disaster which inhibits its capacity to adopt new technology,

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its speed of convergence is correspondingly slower in relation to other countries (including rich countries), thereby producing the phenomenon of transitionally divergent behavior in relation to other countries. In other words, heterogeneous neoclassical economic growth may accommodate a family of potential growth paths in which some divergence may be manifest. If over time the speed of learning in the divergent economies becomes faster than the speed of technology creation in convergent rich economies, there is recovery and catch-up. In this event, the inequality that was initially generated by the divergence becomes transient, and ultimate convergence in world economic growth can be achieved.

Transitional economic behavior of the type described in the last paragraph leads to another major question: what variables govern the behavior of x_{it} and influence its transitional heterogeneity? While this question is not directly addressed in the present paper, the methods developed here for studying empirical economic transitions in growth performance are suited to address similar issues regarding the transition behavior of the factors that influence economic growth.

To accommodate the time series and cross-sectional heterogeneity of technological progress in growth empirics, this paper proposes a new econometric approach based on the analysis of an economy's transition path in conjunction with its growth performance. The transition path can be measured by considering the relative share of per capita log real income of country i in total income, or $h_{it} = \log y_{it}/\overline{\log y_t}$, where $\overline{\log y_t}$ denotes the cross-section average of log per capita real income in the panel or a suitable subgroup of the panel.[1] Under certain regularity conditions on the growth paths, the quantity h_{it} eliminates the common growth components (at least to the first order), and provides a measure of each individual country's share in common growth and technological progress. Moreover, since h_{it} is time dependent, it describes how this share evolves over time, thereby providing a measure of economic transition. In effect, h_{it} is a time-dependent parameter that traces out a *transition curve* for economy i, indicating that economy's share of total income in period t. If there is a common source of sustained economic growth μ_t, then with the diffusion of technology and learning across countries, learning through formal education, and on-the-job learning (Lucas, 2002), we may reasonably suppose that all countries ultimately come to share (to a greater or lesser extent) in this growth experience. In this context, the parameter h_{it} captures individual economic transitions as individual countries experience this phenomenon to varying extents. As with the Galton fallacy, we do not expect all countries to converge. There will always be an empirical distribution of growth and per capita income among nations, as indeed there is between individuals within a country. However, there can still be convergence in the sense of an elimination of divergent behavior (as even the poorest countries begin to catch up) and an ultimate narrowing of the differences. Transitional growth empirics of the type considered in this paper seek to map these differences over time in an orderly manner that provides information about the transition behavior of countries in a world economy as they evolve toward a limit distribution in which all countries share in the common component in economic growth.

The paper is organized as follows. Section 2 outlines some of the stylized facts that have emerged in the economic growth literature and provides some new ways of looking at these regularities. Section 3 studies some of the issues that arise in allowing for heterogeneity in neoclassical growth models and examines links between temporal heterogeneity in the speed of convergence and transitional divergence. Section 4 formalizes the concept of an economy's transition curve, which reveals the extent to which an individual economy shares at each point in

[1] The idea of measuring transitions by means of a transition parameter was first suggested in the working paper Phillips and Sul (2003).

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time in the common growth component of a group of economies. Also, we develop an econometric formulation of this concept, which provides the time profile of transition for one economy relative to a group average. This relative transition curve is identified and can be fitted using various smoothing methods. Use of the transition curve concept is demonstrated, some of its properties are discussed, and a new regression method is given for studying convergence, transition and divergence among economies. Empirical applications of these methods are reported in Section 5, where we study regional transitions in the USA, national economic performance in the OECD nations, and growth and transitional divergence in the world economy using the Penn World Tables (PWT). Some conclusions and prospects for further research are given in Section 6. Supporting technical arguments, a demonstration of bias and inconsistency in augmented Solow regressions under heterogeneous technological progress, and information on the data are given in the Appendix. Further aspects of the econometric methodology used here are discussed in Phillips and Sul (2007).

2. HETEROGENEOUS TECHNOLOGY AND GROWTH

A typical Solow growth model assumes homogeneous technological progress, so that in a cross-section setting all economies experience technological improvements at the same rate over time, while operating from different initial levels. Under such homogeneity in technology, observed cross-section income heterogeneity is difficult to explain, leading researchers to consider more plausible assumptions that allow technological growth rates to differ across countries and over time and to be endogenously determined. For example, Parente and Prescott (1994) introduced an 'adoption barrier' to explain cross-sectional income heterogeneity, Benhabib and Spiegel (1994) specified a model where technology depends on a nation's human capital stock level, and Howitt and Mayer-Foulkes (2005) suggested 'cost of learning' as a mechanism for inducing heterogeneity in technology. According to these theories, a more plausible assumption for empirical work is that the technology growth rates may differ across countries and over time. Some empirical studies have indeed moved in this direction, although they do not fully account for temporal/time-varying heterogeneity. For instance, Islam (1995) allowed for time-invariant individual heterogeneity using fixed effects and Lee *et al*. (1997, 1998) considered time-invariant growth heterogeneity using individual specific slope coefficients.

To account for the temporal and transitional heterogeneity, we introduce time-heterogeneous technology by allowing technological progress, A_{it}, to follow a path of the form $A_{it} = A_{i0} \exp(x_{it}t)$, so that the 'growth rate' parameter x_{it} may differ across countries and over time but may possibly converge to the same rate as $t \to \infty$ either for all countries or for certain groups of countries with a common rate within each group. Under this heterogeneous technology, the individual transition path of log per capita real income, $\log y_{it}$, depends on the technological progress parameter x_{it}, so that $\log y_{it}$ evolves as

$$\log y_{it} = \log \tilde{y}_i^* + [\log \tilde{y}_{i0} - \log \tilde{y}_i^*]e^{-\beta_{it}t} + x_{it}t \tag{3}$$

where β_{it} is a time-varying speed of convergence parameter whose value is given by (see the Appendix for the derivation)

$$\beta_{it} = \beta - \frac{1}{t} \log \left\{ 1 - d_{i1} \int_0^t e^{\beta_p}(x_{ip} - x)\, \mathrm{d}p \right\} \tag{4}$$

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where $d_{i1} = 1/(\log k_{i0} - \log k_i^*)$. Clearly, β_{it} depends on the whole time profile of the rate of technological progress $\{x_{ip}\}_{p \le t}$ since initialization at $t = 0$. Note that in cases where technology is convergent, the deviations $x_{ip} - x$ are bounded and so $\beta_{it} \to \beta > 0$ as $t \to \infty$.[2]

As is apparent from (3) and (23), when $x_{it} = x$, the relative income differential between economies, $(\log y_{it} - \log y_{jt})$, is explained only by the initial real effective per capita income. However, when $x_{it} \neq x$ during transition periods, the relative technological differential between x_{it} and x_{jt} (and the historical trajectory of this differential) also contributes to the income difference. Note that $e^{-\beta_{it}t} \to 0$ as $t \to \infty$, and if the convergence rate of this exponential term in (3) is fast relative to the convergence rate of x_{it}, then the main long-run determinant of the relative income difference in the rates of technological accumulation. In this case, the relative income difference between two economies may be well explained by the relative difference in technology accumulation. For large t, $\log y_{it}$ eventually follows a long-run path determined by the term $x_{it}t$ in (3). Hence analyzing the dynamics of $\log A_{it}$ and the past history of x_{it} are key elements in understanding transitional income dynamics.

3. ECONOMIC TRANSITION CURVES

This section consists of two subsections. We start by developing an econometric formulation of the neoclassical model given earlier that allows for heterogeneity in the speed of convergence and transition effects over time. The new formulation is a nonlinear factor model and involves the product of a time-varying idiosyncratic element (b_{it}), which measures individual transition effects, and a common (stochastic) trend factor (μ_t), which captures the effects of common technology. Next, we suggest a mechanism for measuring the transition effects by means of a transition curve, which may be interpreted as approximating the trajectory over time of the idiosyncratic elements b_{it}. The second subsection applies this model and approach to three panel datasets, which display various forms of transition effects, including some cases of transitional divergence.

3.1. Transition and Relative Transition Curves

It is helpful in this development to use some general specification of the trending mechanism. It is sufficient for our purpose that there be some underlying trend mechanism, which may have both deterministic and stochastic components, and that this trend mechanism be a common element (for instance, arising from knowledge, technology and industry in developed countries) in which individual economies can share. We denote this common trend element by μ_t. The extent to which economies do share in the common trend depends on their individual characteristics and this will ultimately be manifest in their growth performance and the shape of any economic transitions that occur.

From (3), the actual transition path of log per capita real income can be written as follows:

$$\log y_{it} = \log \tilde{y}_i^* + \log A_{i0} + [\log \tilde{y}_{i0} - \log \tilde{y}_i^*]e^{-\beta_{it}t} + x_{it}t = a_{it} + x_{it}t \tag{5}$$

where

$$a_{it} = \log \tilde{y}_i^* + \log A_{i0} + [\log \tilde{y}_{i0} - \log \tilde{y}_i^*]e^{-\beta_{it}t} \tag{6}$$

[2] Note that β_{it} can be expressed as $\beta + O(t^{-1} \log t)$.

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As $t \to \infty$, (6) is a decay model for a_{it} which captures the evolution $a_{it} \to \log \tilde{y}_i^* + \log A_{i0}$. Correspondingly, for large t, $\log y_{it}$ eventually follows a long-run path that is determined by the term $x_{it}t$ in (5).

Following the discussion above, the growth path $x_{it}t$ is presumed to have some elements (and sources) that are common across economies. We use μ_t to represent this common growth component and can think of μ_t as being dependent on a common technology variable which enters as a factor of production for each individual economy. According to this view, all economies share to a greater or lesser extent in certain elements that promote growth—such as the industrial and scientific revolutions and internet technology. We may then write (5) in the following form:

$$\log y_{it} = \left(\frac{a_{it} + x_{it}t}{\mu_t} \right) \mu_t = b_{it}\mu_t \tag{7}$$

where b_{it} explicitly measures the share of the common trend μ_t that economy i experiences. In general, the coefficient b_{it} measures the transition path of an economy to the common steady-state growth path determined by μ_t. During transition, b_{it} depends on the speed of convergence parameter β_{it}, the rate of technical progress parameter x_{it} and the initial technical endowment and steady-state levels through the parameter a_{it}.

Note that growth convergence requires the following condition:

$$x_{it} \to x, \text{ for all } i \text{ as } t \to \infty \tag{8}$$

Condition (8) is sufficient for the convergence of the growth rate of $\log y_{it}$, but only necessary for the level convergence of $\log y_{it}$.

In a neoclassical growth framework, steady state common growth for $\log y_{it}$ may be represented in terms of a simple linear deterministic trend $\mu_t = t$. Then, according to (7), $b_{it} = x_{it} + a_{it}/t$ and, under the growth convergence condition (8), we have the convergence $b_{it} \to x$ as $t \to \infty$. Further, when the economies have heterogeneous technology and x_{it} converges to x_i, we have

$$b_{it} = x_{it} + \frac{a_{it}}{t} \to x_i, \text{ as } t \to \infty \tag{9}$$

so that x_i determines the growth rate of economy i in the steady state. The quantity b_{it} therefore plays a key role as a transition parameter in this framework.

In more general models and in empirical applications, the common growth component μ_t may be expected to have both deterministic and stochastic elements, such as a unit root stochastic trend with drift. In the latter example, μ_t is still dominated by a linear trend asymptotically and conditions like (9) then hold as limits in probability. While this case covers most practical applications, we may sometimes want to allow for formulations of the common growth path μ_t that differ from a linear trend even asymptotically. Furthermore, a general specification allows for the possibility that some economies may diverge from the growth path μ_t, while others may converge to it. These extensions involve some technical complications that can be accommodated by allowing the functions to be regularly varying at infinity (that is, they behave asymptotically like power functions). We may also allow for individual country standardizations for log per capita income, so that expansion rates may differ, as well as imposing a common standardization for μ_t. Details of such extensions of the present setup will be reported elsewhere.

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The estimation of b_{it} is not possible without imposing some smoothness or structural restrictions since the total number of unknowns is the same as the number of observations. Parametric assumptions enable the time profile of b_{it} to be fitted by filtering methods such as the Kalman filter. Smoothness conditions and deterministic assumptions allow for nonparametric estimation by kernel methods or sieve techniques. However, such methods are presently not well developed for fitting stochastic processes, rather than deterministic functions.

An alternative approach to modeling the transition elements b_{it} that is convenient in the present context is to construct the following relative transition coefficient:

$$h_{it} = \frac{\log y_{it}}{N^{-1}\sum_{i=1}^{N} \log y_{it}} = \frac{b_{it}}{N^{-1}\sum_{i=1}^{N} b_{it}} \tag{10}$$

which eliminates the common growth component by scaling and measures the transition element for economy i relative to the cross-section average. The variable h_{it} traces out an individual trajectory for each i relative to the average, so we call h_{it} the 'relative transition path.' At the same time, h_{it} measures economy i's relative departure from the common steady-state growth path μ_t. Thus, any divergences from μ_t are reflected in the transition paths h_{it}. While many paths are possible, a case of particular interest and empirical importance occurs when an economy slips behind in the growth tables and diverges from others in the group. We may then use the transition path to measure the extent of the divergent behavior and to assess whether or not the divergence is transient.

When there is a common (limiting) transition behavior across economies, we have $h_{it} = h_t$ across i, and when there is ultimate growth convergence we have

$$h_{it} \to 1, \quad \text{for all } i, \quad \text{as } t \to \infty \tag{11}$$

This framework of growth convergence admits a family of relative transitions, where the curves traced out by h_{it} may differ across i in the short run, while allowing for ultimate convergence when (11) holds in the long run. Removing the common (steady-state) trend function μ_t, Figure 1 shows some examples of relative transition paths, each satisfying the growth convergence condition (11).

While the criterion for the ultimate convergence of economy i to the steady state is given by (11), the manner of economic transition and convergence can be very different across economies. Figure 1 shows three different stylized paths. Economies 2 and 3 have quite different initializations and their transitions also differ. While both relative transition parameters converge monotonically to unity, path 3 involves transition from a high initial state, typical of an already advanced industrial economy, whereas path 2 involves transition from a low initial state that is typical of a newly industrialized and fast-growing economy. Economy 1, on the other hand, has the same initialization as 2 but its relative transition involves an initial phase of divergence from the group, followed by a catch-up period, and later convergence. Such a transition is typical of a developing country that grows slowly in an initial phase (transition phase A), begins to turn its economic performance around (phase B) and then catches up and converges (phase C).

Also, as Figure 1 illustrates in a stylized way, when there is temporal and cross-section heterogeneity, there exists an infinite number of possible transition paths, some showing periods of transitional divergence (such as economy 1) even in cases where there is ultimate convergence.

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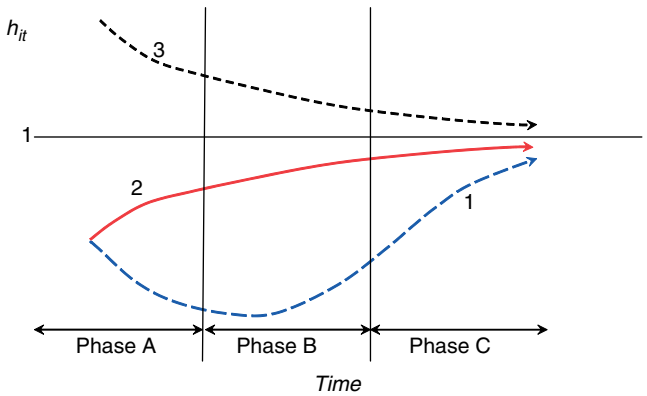


Figure 1. Relative transition curves h_{it} and phases of transition. This figure is available in color online at www.interscience.wiley.com/journal/jae

3.2. Transition Phases and Divergence: Some Graphical Illustrations

As suggested in this stylized diagram, diverse patterns of economic transition are possible when we allow for cross-sectional and time series heterogeneity in the parameters of a neoclassical growth model. This potential for diversity in transition is illustrated in the following empirical examples involving regional and national economic growth. Similar panel datasets to those used here have been extensively analyzed in the growth convergence literature in the past. Our application now focuses attention on the phenomenon of economic transition as part of a larger empirical story regarding convergence and divergence issues. We start by providing some graphical illustrations of the various phases of transition in the empirical data and then proceed to conduct some formal statistical tests based on empirical (decay model) regressions.

The first illustration is based on regional economic growth among the 48 contiguous US states.[3] In this example, there is reasonable prior support for a common rate of technological progress and ultimate growth convergence but we may well expect appreciable heterogeneity across states in the transition paths. Figure 2 displays the relative transition parameters calculated for log per capita income in the 48 states over the period from 1929 to 1998 after eliminating business cycle components.[4] Evidently, there is heterogeneity across states, but also a marked reduction in dispersion of the transition curves over this period, together with some clear evidence that the relative transition curves narrow towards unity, as indicated in the convergence criterion (11).

The second illustration involves a panel of real per capita income for 18 western OECD countries taken from the OECD historical dataset. Panel A in Figure 3 displays the relative transition parameters for log per capita income in these 18 OECD countries[5] between 1929 and 2001. The countries were selected on the basis of data availability and are listed in the Data Appendix. The observed time profiles of transition for these OECD nations are quite different from those of Figure 2, even though the time frame is similar. For the OECD nations, the relative

[3] The data source for US state per capita real income is the Bureau of Economic Analysis.
[4] Both HP and trend regression methods were used to eliminate business cycle components and the results showed little sensitivity to the method employed.
[5] The pattern of the relative transition curves for a restricted panel of 14 European countries is very similar to that shown in Figure 3, Panel A.

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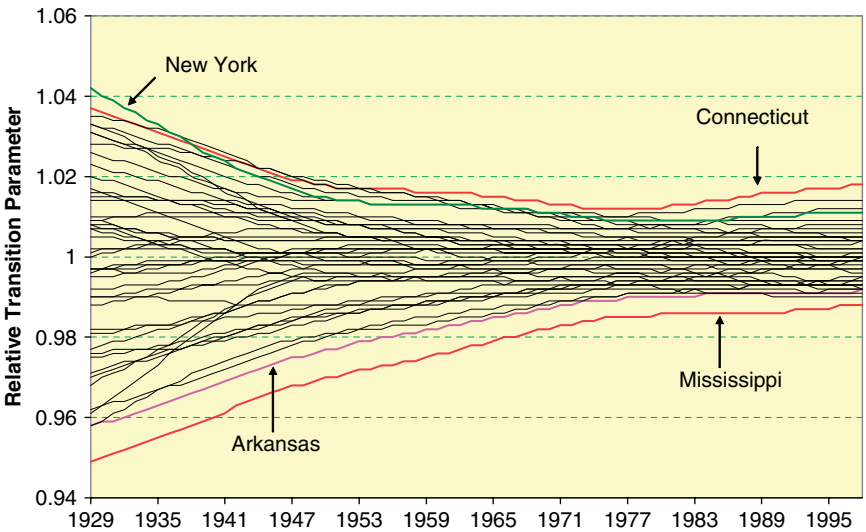


Figure 2. Transition paths for the 48 contiguous US states. This figure is available in color online at www.interscience.wiley.com/journal/jae

transition parameters initially seem to display no coherent pattern and, in some cases, even appear to diverge before World War II. Around 1950, however, the pattern of transition appears to change and subsequently becomes similar to that of Figure 2. Over the latter part of the period, there is a noticeable narrowing in the transition curves towards unity, indicating a clear tendency to converge towards the end of the period.

Panels B and C in Figure 3 show the relative transition curves for certain subgroups of countries against the benchmark of the USA.[6] We have created five economic subgroups in this exercise. Except for the former UK colonies, all subgroups show clear evidence of some transitional divergence with a turn-around by the end of World War II. After that, all of these subgroups reveal a strong tendency towards convergence with the USA. Evidently, panel B in Figure 3 provides an empirical illustration of the stylized patterns of economic performance characterized as phases B and C of Figure 1. Extending the panel back to 1870 and through to 1930, panel C in Figure 3 shows transition curves that are similar in form to phase A (transitional divergence) in the stylized patterns of Figure 1, with evidence of the phase B turn-around coming towards the end of this period.

The final illustration is based on log per capita income in 98 PWT countries in the world economy over various periods. The country selection is mainly based on data availability. Given the large number of countries and the wide variation in the data, it is helpful to take subgroup averages to reduce the number of transition curves, which we show against the benchmark of the

[6] For all subgroups, the USA enters as the numeraire country. The relative transition curves for subgroup k are calculated as follows: Let N_k be the total number of countries in the subgroup. Since the USA is always included in the subgroup, the total number of countries excluding the USA is $N_k - 1$. First we calculate $h_{it} = \log\ y_{it} \left(\left[N_k^{-1} \sum_{i=1}^{N_k} \log\ y_{it} \right] \right)^{-1}$ for $i = 1, \ldots, N_k$. Next, we take the cross-sectional average of the h_{it} excluding the USA. That is, $h_{kt} = \frac{1}{N_k - 1} \sum_{i \neq \text{USA}}^{N_k} h_{it}$.

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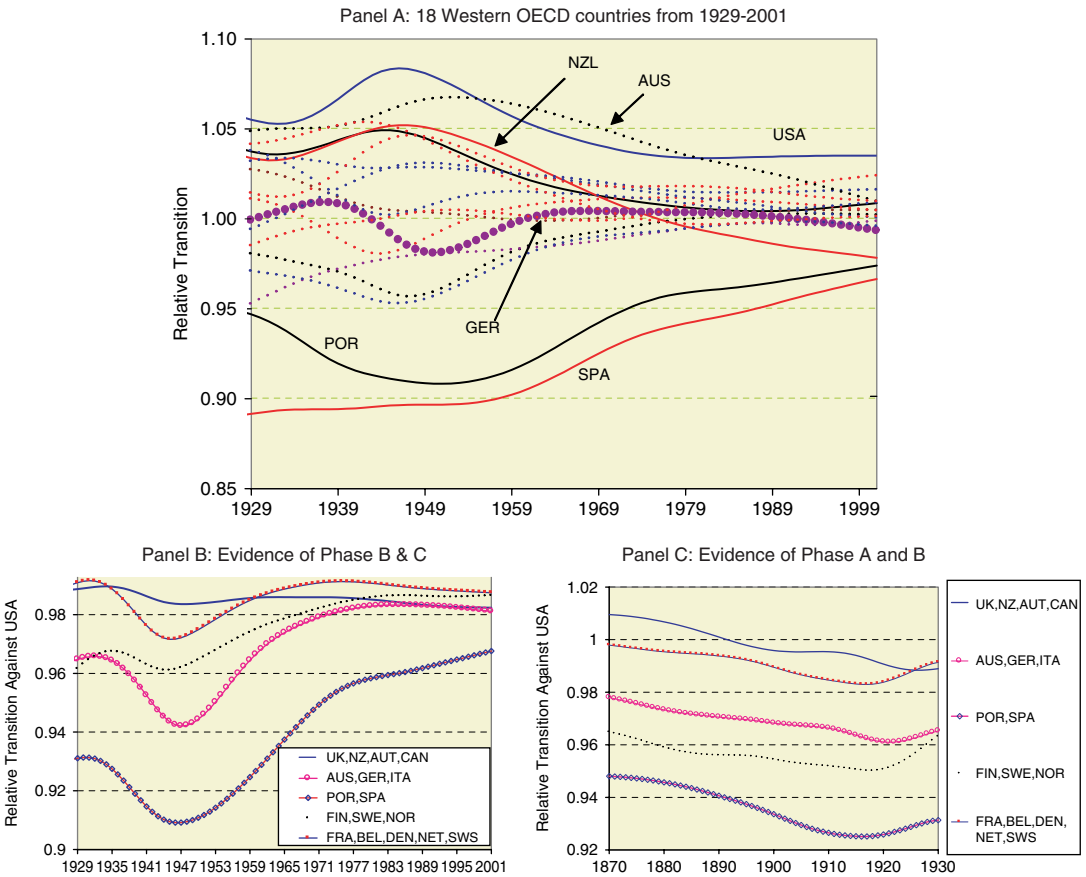


Figure 3. OECD transition paths. This figure is available in color online at www.interscience.wiley.com/journal/jae

19 OECD countries.[7] The subgroups are based on total population and geographical region. Phase A transitions are found in two of these subgroups: the countries of sub-Saharan Africa, and the Latin American and Caribbean economies from 1960 to 2003 (panel 3 in Figure 4). Phases B and C occurred in three cases—India, China and Korea—over the period from 1953 to 2003 (panel 1 in Figure 4). Finally, phase C transitions are evident in two subgroups: the Asian dragons and the newly industrialized economies (NIEs) from 1960 to 2003 shown in panel 2 in Figure 4.

From these findings about the present standing of these economic groups and assuming that the world economies are in transition to ultimate convergence on a path that is related to long-run historical OECD growth, then we can expect that China and India will continue to grow faster over the next decade than the OECD nations as they experience phase C transition; and, sooner or later, we might expect to witness the sub-Saharan and Latin American countries entering phase B

[7] The relative transition curves are calculated in a similar fashion to that described in footnote 11. The only difference is that instead of the USA as numeraire, the 19 OECD countries are used as numeraire countries. That is, the cross-sectional average of h_{it} is given by $h_{kt} = \frac{1}{N_k - 19} \sum_{i \neq \text{OECD}}^{N_k} h_{it}$.

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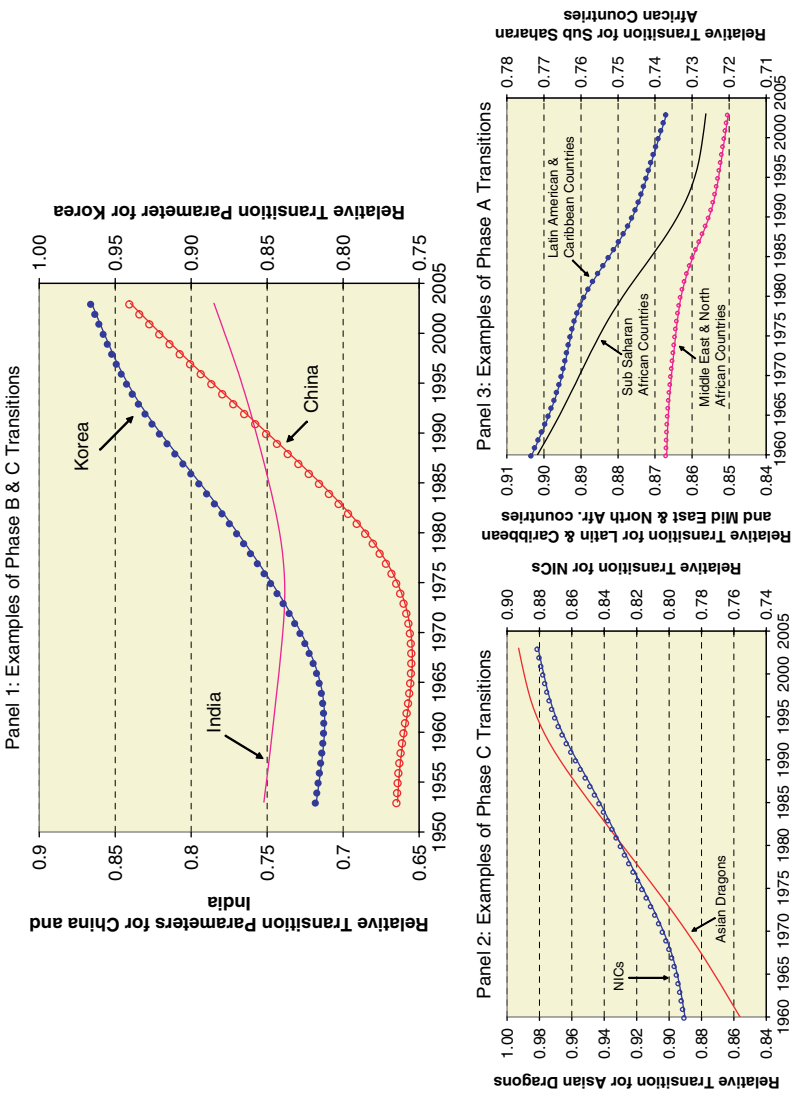


Figure 4. World transition paths. This figure is available in color online at www.interscience.wiley.com/journal/jae

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transition when they begin to turn around their economic performance and start to catch up with the 19 OECD countries. However, from the evidence to date in these figures, we cannot distinguish for the sub-Saharan Africa and Latin American countries whether or when such changes may occur.

We now provide some formal econometric procedures for evaluating transition curves to shed light on growth convergence and convergence clustering issues.

4. TESTING GROWTH CONVERGENCE

The conventional conditional β-convergence test is designed to assess whether the speed of convergence parameter (β) in (23) is positive or negative. This simple and rather intuitive test works under homogeneity of technology progress and has been widely applied. When technology is heterogeneous across countries, allowing transition periods of the type just described, then the speed of convergence β_{it} is time varying and depends on the time profile of the rate of technological progress $\{x_{is}\}_{s\leq t}$. This dependence is a considerable complication that affects the properties of the regression equation that is traditionally used in tests that focus attention on β. Under heterogeneity, as we have seen, growth convergence depends on whether $x_{it} \to x$, for all i, as in (8), or whether $b_{it} \to b$ and $h_{it} \to 1$, for all i, as in (11). This condition implies the relative convergence studied in Phillips and Sul (2007). Relative convergence is defined as

$$\lim_{t\to\infty} \frac{\log y_{it}}{\log y_{jt}} = 1 \text{ for all } i \text{ and } j$$

Relative convergence in discrete time series implies growth convergence in the long run rather than level convergence. Moreover, the relative convergence concept holds when the common component μ_t in (7) follows either a nonstationary stochastic trend with drift or a trend stationary process. In either case, the common component diverges at $O_p(t)$ rate. Hence, if b_{it} converges faster than the $O_p(t)$ divergence rate, then relative convergence implies absolute or level convergence. When the convergence rate of b_{it} is slower than the divergence rate of μ_t, relative convergence occurs but absolute convergence fails.

4.1. Pitfalls of Existing Convergence Tests

First, we briefly discuss some issues that arise with other approaches when there is heterogeneous transition. Perhaps the most popular approach in empirical work is the augmented Solow regression (Barro and Sala-i-Martin, 1992; Mankiw *et al.*, 1992), which takes the general form

$$\log\left(\frac{y_{it}+1}{y_{it}}\right) = a_0 + a_1 \log y_{it} + \mathbf{z}_i' a_2 + e_i \tag{12}$$

where $\mathbf{z}_i$ is a vector of proxy (determining) variables for the steady-state log levels $\log \tilde{y}_i^*$. When there is heterogeneity in technological progress, the parameter β_{it} is time and country varying, and this materially affects the specification (12) in important ways. As shown in Phillips and Sul (2005), the coefficient a_1 in (12) is then a function of β_{it} and x_{it}. In particular, if we assume that the β_{it} are independently drawn from some distribution across economies, then a_1 can be written in terms of the moment-generating function of that distribution, so that for some given t we have

$$a_1 = -(1 - E[e^{-\beta_i^+}]), \quad \text{where } \beta_i^+ = \beta_{it+1} + (\beta_{it+1} - \beta_{it})t, \tag{13}$$

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and where β_{it} is defined in (4). The residual is absorbed into the error e_i in (12). Moreover, as shown in Phillips and Sul (2005), under heterogeneous technological progress, the regression error e_i includes terms that involve technology levels $\log A_{it}, \log A_{it+1}$ and variables that are correlated with $\log y_{it}$ and $\mathbf{z}_i$. Hence, least squares regression on (12) is inconsistent and the bias arises from two separate sources: omitted variables and endogeneity. The main issue regarding the use of augmented Solow regression such as (12) in practice is that, under conditions of transitional heterogeneity, estimation of the coefficient a_1 is biased and inconsistent, so negative estimates of a_1 cannot be directly interpreted as evidence of growth convergence. Moreover, $a_1 = -(1 - E[e^{-\beta_i^+}]) < 0$ does not imply that $x_{it} = x$ or that $x_{it} \to x$ as $t \to \infty$, so that it is possible to get negative estimates of a_1 even under conditions of divergence. A further issue in the use of augmented Solow regressions is the inconsistency of estimates of the coefficient vector a_2 of the proxy variables in (12). Again, due to omitted variable and endogeneity bias, estimates of a_2 may suffer a reversal of sign, leading to false inferences concerning the direction of influence of certain growth determinants. This issue is being investigated more fully in other work.

Another common approach to growth convergence analysis involves the use of cointegration and unit root tests. The conventional cointegration test approach is based on examining the time series properties of log-level inter-economy differentials such as $\log y_{it} - \log y_{jt}$. In particular, such series can be tested for the presence of a stochastic trend by means of an empirical regression of the form

$$\log y_{it} - \log y_{jt} = c_1 + c_2 t + \rho\ (\log y_{it-1} - \log y_{jt-1}) + \text{error} \tag{14}$$

Bernard and Durlauf (1995, 1996) and Evans and Karras (1996) omit the linear trend term in this regression and test whether $\rho < 1$ (convergence) or $\rho = 1$ (divergence). If $c_1 \neq 0$ but $\rho < 1$, Evans and Karras (1996) interpret this as conditional convergence. Carlino and Mills (1993) include the linear trend term in (14) and test whether $c_2 = 0$ and $\rho < 1$. When the model involves heterogeneous transition, we may write the inter-economy differentials as

$$\log y_{it} - \log y_{jt} = (a_{it} - a_{jt}) + (x_{it} - x_{jt})\ t = (b_{it} - b_{jt})\mu_t$$

using (5) and (7). Suppose the common growth component μ_t follows a stochastic trend with drift, so that $\mu_t = \mu_0 + \mu t + \sum_{s=1}^{t} \zeta_s$ for some stationary process ζ_s and initialization μ_0. Then

$$\log y_{it} - \log y_{jt} = (b_{it} - b_{jt})(\mu_0 + \mu t) + (b_{it} - b_{jt}) \sum_{s=1}^{t} \zeta_s$$

and $\log y_{it} - \log y_{jt}$ has a unit root component for all t. Existence of a unit root in the differential $\log y_{it} - \log y_{jt}$ does not necessarily imply growth divergence because it is possible that the convergence condition $b_{it} - b_{jt} \to_p 0$ still holds.[8] Hence, depending on the stochastic order of the idiosyncratic factor differential $b_{it} - b_{jt}$, cointegration type tests may or may not reveal whether growth convergence applies.[9]

[8] In that event, we have $\log y_{it} - \log y_{jt} = O_p(\mu(b_{it} - b_{jt})\ t + (b_{it} - b_{jt})\ t^{1/2}) = O_p((b_{it} - b_{jt})t^{1/2})$, with the second equality holding when $\mu = 0$ and there is no drift.

[9] When b_{it} and b_{jt} converge to some common b as $t \to \infty$, $\log y_{it}$ is 'asymptotically' cointegrated with $\log y_{jt}$. Even in this case, when the speed of the convergence of b_{it} is slower than the speed of the divergence of μ_t, the residual $(b_{it} - b_{jt})\mu_t$ contains nonstationary characteristics. Therefore, conventional cointegration tests will typically have low power in detecting the asymptotic co-movement. See Phillips and Sul (2007) for more discussion on this issue.

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Lastly, many empirical studies have considered tests associated with the notion of σ-convergence. For example, Evans (1996) examined the time-varying pattern of the cross-sectional variance of $\log y_{it}$. Under temporal and transitional heterogeneity in (7), the cross-sectional variance, V_t, can be expressed as

$$V_t = \frac{1}{N}\sum_{i=1}^{N}\left(\log y_{it} - \frac{1}{N}\sum_{i=1}^{N}\log y_{it}\right)^2 = \sigma_{bNt}^2\mu_t^2$$

where $\sigma_{bNt}^2 = \frac{1}{N}\sum_{i=1}^{N}\left(b_{it} - \frac{1}{N}\sum_{i=1}^{N} b_{it}\right)^2$. Evans (1996) claimed that under growth convergence the cross-sectional variance, V_t, should be stationary. However, as indicated above, V_t can manifest nonstationary characteristics depending on whether the convergence rate of b_{it} is slower or faster than the divergence rate of μ_t.

4.2. The Log *t* Convergence Test

Since growth convergence under transitional heterogeneity depends explicitly on the condition $b_{it} - b_{jt} \to_p 0$, it seems appropriate to develop an approach which focuses attention on this condition or some convenient version of it. A particularly simple approach is to work with the relative transition coefficients h_{it}, rather than the coefficients b_{it}, because h_{it} may be directly measured from the data in some cases, as in (10), or is easily computed in others where some prefiltering (e.g., business cycle removal) is performed. In both cases, the common growth component is eliminated. Also, when there is ultimate growth convergence, we have the limit $h_{it} \to_p 1$ for all i as $t \to \infty$, and then the mean square transition differential

$$H_t = N^{-1}\sum_{i=1}^{N}(h_{it} - 1)^2$$

provides a quadratic distance measure for the panel from the common limit. Under convergence, the distance $H_t \to 0$ as $t \to \infty$. When convergence does not hold, the distance remains positive as $t \to \infty$. There are various possibilities: H_t may converge to a non-zero constant, it may remain bounded above zero but not converge, or it may diverge. In the case where there is club convergence, H_t typically converges to a positive constant. With limited time series evidence, it is naturally difficult to distinguish whether H_t converges to zero or a positive constant. To sharpen the distinction and assist in empirical testing it is helpful to use a specific model of transition that is conducive to econometric testing.

In related work (Phillips and Sul, 2007) we have developed such a model and testing procedure. The test is based on a simple time series regression and involves a one-sided t-test of the null hypothesis of convergence against alternatives which include no convergence and partial convergence among subgroups. The test is called the '$\log t$' convergence test because the t-statistic refers to the coefficient of the $\log t$ regressor in the regression equation—see (18) below. What follows is a brief outline of the model and the econometric testing procedure.

To formulate a null hypothesis of growth convergence, we use a semiparametric model for the transition coefficients that allows for heterogeneity over time and across individuals as

$$b_{it} = b_i + \frac{\sigma_i \xi_{it}}{L\,(t)\,t^{\alpha}} \tag{15}$$

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where b_i is fixed, ξ_{it} is i.i.d. (0,1) across i but may be weakly dependent over t, and $L(t)$ is a slowly varying function (like $\log t$) for which $L(t) \to \infty$ as $t \to \infty$. The parameter α governs the rate at which the cross-section variation over the transitions decays to zero over time. This formulation ensures that b_{it} converges to b_i for all $\alpha \geq 0$, which therefore becomes a null hypothesis of interest. If this null holds and $b_i = b_j$ for $i \neq j$, the model allows for transitional periods in which $b_{it} \neq b_{jt}$, thereby incorporating the possibility of transitional heterogeneity or even transitional divergence across i. As shown in Phillips and Sul (2007), further heterogeneity may be introduced by allowing the decay rate α and slowly varying function $L(t)$ to be individual specific without affecting the manner in which the test is conducted. Some regularity conditions on the idiosyncratic scale parameters σ_i and the random variables ξ_{it} are required in order to develop rigorous asymptotics for the regression and these are detailed in Phillips and Sul (2007).

The null hypothesis of convergence may be written as[10]

$$H_0 : b_i = b \ \& \ \alpha \geq 0 \tag{16}$$

which involves the weak inequality $\alpha \geq 0$, since

$$\lim_{t\to\infty} b_{it} = b \quad \text{iff} \quad b_i = b \text{ and } \alpha \geq 0$$
$$\lim_{t\to\infty} b_{it} \neq b \quad \text{iff} \quad b_i \neq b \text{ and } \alpha < 0$$

The alternative hypothesis is given by

$$H_A: \{b_i = b \text{ for all } i \text{ with } \alpha < 0\} \text{ or } \{b_i \neq b \text{ for some } i \text{ with } \alpha \geq 0, \text{ or } \alpha < 0\}$$

One role of the slowly varying component $L(t)$ in (15) is to ensure that convergence holds even when $\alpha = 0$, although possibly at a very slow rate. This formulation accommodates some interesting empirical possibilities where there is slow transition and slow convergence. The alternative hypothesis includes straightforward divergence but more importantly also includes the possibility of club convergence. For example, if there are two convergent clubs, then the alternative can include such a case in which

$$H_A : b_{it} \to \begin{cases} b_1 \text{ and } \alpha \geq 0 & \text{if } i \in G_1 \\ b_2 \text{ and } \alpha \geq 0 & \text{if } i \in G_2 \end{cases}$$

where the number of individuals in G_1 and G_2 aggregates to N. For some b_1 and b_2, so that

$$b_1 = \lim_{N\to\infty} N_1^{-1} \sum_{i\in G_1} b_{it}, \quad b_2 = \lim_{N\to\infty} N_2^{-1} \sum_{i\in G_2} b_{it}$$

[10] It is worth noting that the null hypothesis implies 'relative' convergence, which can be defined as $\lim_{t\to\infty} (\log y_{it} / \log y_{jt}) = 1$. Bernard and Durlauf (1996) and Evans and Karras (1996) consider level convergence, which is defined as $\lim_{t\to\infty} (\log y_{it} - \log y_{jt}) = 0$. To see the difference between the two definitions, let $\mu_t = t$, and consider a case such that $\log y_{1t} = (1 + t^{-\alpha})t$ for $\alpha \geq 0$, and $\log y_{2t} = t$, so that $b_{1t} = 1 + t^{-\alpha}$ and $b_{2t} = 1$. Then $\log y_{1t} - \log y_{2t} = t^{1-\alpha}$, which diverges to positive infinity if $0 \leq \alpha < 1$, while $\log y_{1t} / \log y_{2t} = 1 + t^{-\alpha}$ converges. In discrete time series, relative convergence implies so-called 'growth convergence'. In the above example, we have $\Delta \log y_{1t} = 1 + t^{1-\alpha} - (t - 1)^{1-\alpha}$, $\Delta \log y_{2t} = 1$, so that $\lim_{t\to\infty} (\Delta \log y_{1t} - \Delta \log y_{2t}) = \lim_{t\to\infty} [t^{1-\alpha} - (t-1)^{1-\alpha}] = 0$ for $0 < \alpha < 1$. Also note that the null hypothesis nests level convergence.

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and

$$h_{it} = \frac{b_{it}}{N^{-1}\sum_i b_{it}} \to \begin{cases} \dfrac{b_1}{\lambda b_1 + (1-\lambda)b_2} & i \in G_1 \\ \dfrac{b_2}{\lambda b_1 + (1-\lambda)b_2} & i \in G_2 \end{cases}$$

then

$$H_t = N^{-1}\sum_{i=1}^{N}(h_{it}-1)^2 \to \frac{\lambda(1-\lambda)\{\lambda b_1^2 + (1-\lambda)\, b_2^2\}}{\{\lambda b_1 + (1-\lambda)\, b_2\}^2}$$

for all $\lambda \neq 0$, 1 and $b_1 \neq b_2$. A similar weighted limit is obtained in the case of multiple clubs.

Corresponding to the decay model (15) and under growth convergence the transition distance H_t is shown in Phillips and Sul (2007) to have the limiting form

$$H_t \sim \frac{A}{L(t)^2 t^{2\alpha}} \text{ as } t \to \infty \tag{17}$$

for some constant $A > 0$. Setting $L(t) = \log t$, this formulation leads to the following '$\log t$' regression model:

$$\log \frac{H_1}{H_t} - 2\log(\log t) = a + \gamma \log t + u_t, \quad \text{for } t = T_0, \ldots, T \tag{18}$$

where $H_t = N^{-1}\sum_{i=1}^{N}(h_{it}-1)^2$ and $h_{it} = \log y_{it}/N^{-1}\sum_{i=1}^{N}\log y_{it}$. In (18), the initial observation in the regression is $T_0 = [rT]$ for some $r > 0$, so that empirical $\log t$ regressions are based on time series data in which the first r% of the data is discarded.[11] The second term on the left-hand side in (18), $-2\log(\log t)$, plays the role of a penalty function and improves test performance particularly under the alternative. For instance, under the alternative of club convergence, the transition distance H_t converges to a positive quantity as $t \to \infty$ and the inclusion of the penalty term in the regression gives the test discriminatory power between overall convergence and club convergence.

Under the null of growth convergence, the point estimate of the parameter γ converges in probability to the scaled speed of convergence parameter 2α. The corresponding t-statistic in the regression is constructed in the usual way using HAC standard errors. This t-statistic diverges to positive infinity when $\alpha > 0$ and converges weakly to a standard normal distribution when $\alpha = 0$. The convergence test then proceeds as a one-sided t-test of $\alpha \geq 0$. Under the alternative of growth divergence or club convergence, the point estimate of γ converges to zero regardless of the true value of α, but its t-statistic diverges to negative infinity, thereby giving the one-sided t-test discriminatory power against these alternatives.

In conducting the regression (18) we may employ either h_{it} computed directly from $\log y_{it}$ as in (10) or, as discussed above, use filtered data $\hat{h}_{it}$ based on fitted values $\widehat{\log y_{it}}$ from a coordinate trend regression (Phillips, 2005) or a smoothing filter that removes business cycles.

We are interested not only in the sign of the coefficient $\gamma = 2\alpha$ of $\log t$ in (18) but also its magnitude, which measures the speed of convergence of b_{it}.[12] Hence if $\gamma \geq 2$ (i.e., $\alpha \geq 1$) and

[11] Phillips and Sul (2007) suggest r values in the interval [0.2, 03]. This data trimming focuses attention on the latter part of the sample data, validates the regression equation in terms of the asymptotic representation of the transition distance (17), and ensures test consistency in growth convergence applications.

[12] In particular, under (15) we have $b_{it} = b_i + O_p\left(\frac{1}{L(t)t^{\alpha}}\right)$.

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the common growth component μ_t follows a random walk with drift or a trend stationary process, then values of γ that are this large will imply convergence in level per capita incomes. Meanwhile, if $2 > \gamma \geq 0$, then this speed of convergence corresponds to conditional convergence, i.e., income growth rates converge over time.

4.3. Empirical Evidence

We employ the log t-test with four panels. The first panel consists of income data for the 48 contiguous US states from 1929 to 1998; the second panel consists of 18 Western OECD countries from 1870 to 2001; the third panel consists of 152 PWT countries from 1970 to 2003.

Table I reports the results of applying the log t-test with three panel datasets. The last two columns in the table report the point estimates γ and their standard errors, which are computed using an automated HAC procedure. For the state income data, the estimate $\hat{\gamma} = 0.46$ is significantly positive, so that there is strong evidence in support of H_0 in (16) and for convergence as defined in (11). If the common stochastic trend component follows either a random walk with a drift or a trend stationary process, then the speed of convergence parameter is significantly below 2, so that absolute level convergence can be rejected.

The second panel involves log per capita GDP for 18 Western OECD countries. The point estimate of $\hat{\gamma} = 1.71$ is large for the period 1870–2001, so that the null hypothesis of convergence is strongly accepted, but $\hat{\gamma}$ is also significantly below 2, so that the hypothesis of absolute level convergence is rejected. In order to analyze the impact of transition on the log t-test, we did some subsample analyses. The first subsample covers the period 1870–1929 and can be characterized as phase A transition as discussed in the context of Figure 7. During this period, we find that $\hat{\gamma} = -0.42$, which implies divergence. The second subsample covers 1911–1970 and this period has phase B (or transitional divergence and turn around) characteristics. For this period, the point estimate is $\hat{\gamma} = -0.11$ and is not significantly different from zero. The last subsample covers 1940–2001 and for this period the point estimate is $\hat{\gamma} = 1.14$, confirming convergence. The point estimate of the convergence parameter increases as the initialization period is moved forward and as later observations are included.

For the third panel, there is no evidence of convergence: the point estimates, $\hat{\gamma}$, are significantly less than zero for all cases and the estimated standard errors are so small for these large cross-section groups that the null hypothesis of convergence is rejected even at the 0.1% level. This confirms earlier findings.

Table I. Convergence tests

Cases	Time	$\hat{\gamma}$	SE($\hat{\gamma}$)
48 US states	1929–1998	0.46	0.04
18 Western OECD	1870–2001	1.71	0.07
	1870–1929	−0.42*	0.04
	1911–1970	−0.11	0.09
	1940–2001	1.14	0.01
152 PWT	1970–2003	−0.88*	0.01

Note: * Significant at the 5% level.

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5. GROWTH CONVERGENCE CLUBS AND ECONOMIC TRANSITION

The log t regression test has power against cases of club convergence, so we can expect the null hypothesis of convergence to be rejected for data in which there is evidence of club convergence. Accordingly, we investigate the possibility of a club convergence pattern among the 152 PWT countries where overall convergence is firmly rejected. To do so, we utilize a clustering mechanism test procedure which relies on the following stepwise and cross-section recursive application of log t regression tests. The procedure facilitates the empirical discovery of club convergence clusters, using the fact that the log t regression test has power against club convergence alternatives.

5.1. Clustering Algorithm and Convergence Club Empirics

A detailed analysis of the clustering procedure is given in Phillips and Sul (2007). The constructive steps for implementing the procedure are briefly summarized as follows:

- *Step 1 (cross-section ordering)*: Order the countries according to either the amount of final period income or the average of the last half period of incomes.[13]
- *Step 2 (from a core primary group of k^* countries)*: Selecting the first k highest individuals in the panel to form the subgroup G_k for some $2 \leq k < N$, run the log t regression and calculate the convergence test statistic $t_k = t(G_k)$ for this subgroup. Choose the core group size k^* by maximizing t_k over k according to the criterion.[14]

$$k^* = \arg\max_k \{t_k\} \text{ subject to } \min\{t_k\} > -1.65 \tag{19}$$

  If the condition $\min\{t_k\} > -1.65$ does not hold for $k = 2$, then the highest individual in G_k can be dropped from each subgroup and new subgroups $G_{2j} = \{2, .., j\}$ formed for $3 \leq j \leq N$. The step can be repeated with test statistics $t_j = t(G_{2j})$.[15]
- *Step 3 (sieve the data for new club members)*: Add one country at a time to the core primary group with k^* members and run the log t-test again. Include the new country in the convergence club if the associated t-statistic is greater than the criterion c^*.
- *Step 4 (recursion and stopping rule)*: Form a second group from those countries for which the sieve condition fails in Step 3. Run the log t-test to see if $t_{\hat{\gamma}} > -1.65$ on this group, i.e., if this group satisfies the convergence test. If so, conclude that there are two convergence club groups: the core primary group and the second group. If not, repeat step 1 through step 3 to see if this second group can itself be subdivided into convergence clusters. If there is no k in Step 2 for which $t_k > -1.65$, conclude that the remaining countries do not contain a convergence subgroup and so the remaining countries have divergent behavior.

[13] We note that De Long (1988) criticized the *ex post* sample selection method used in Baumol (1986) to study productivity growth and convergence in OECD countries. The clustering results here instead use a data-based approach to determine convergence groupings among countries. The first step in the procedure utilizes an initial cross-section ordering according to final period income (or the average last half period income) and in this sense might be critiqued as using *ex post* classification. However, the critical step in the procedure is the selection of a core group of countries central to each cluster and this determination is made on a conservative data-based approach and is robust to initial data orderings.

[14] The condition $\min\{t_k\} > -1.65$ plays a key role in ensuring that the null hypothesis of convergence is supported for each k.

[15] If the condition $\min\{t_j\} > -1.65$ is not satisfied for the first $j = 2$, the step may be repeated again, dropping the highest individuals in G_j and proceeding as before. If the condition does not hold for all such sequential pairs, then we conclude that there are no convergence subgroups in the panel.

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In this procedure, the ordering of the cross-sectional individuals is not as important as the determination of the core group, which plays a key role in the clustering algorithm. The core group should not include erroneous members and this is why the core group size is chosen by maximizing the convergence test statistic t_k over k. In the third step, the choice of the sieve criterion c^* is associated with the desired degree of conservativeness in the clustering method. Higher c^* implies less risk of including a wrong member of the convergence club. As c^* approaches zero (from below), the sieve condition becomes more conservative. When T is small, the sieve criterion c^* can be set to zero to ensure that it is highly conservative, whereas for large T, c^* can be set to the asymptotic 5% critical value -1.65. One consequence of extremely conservative testing induced by setting $c^* = 0$ is that it tends to raise the chance of finding more convergent clubs than the true number. To avoid such overdetermination, we may run log t regression tests across the subgroups to assess evidence in support of merging clubs into larger clubs.

Table II shows the results of applying these clustering procedures to the PWT data involving 152 countries over the period 1970–2003. The table has three panels. The left panel (headed 'Initial classification') shows the empirical results obtained from a direct application of the clustering algorithm described above. The second column under this heading reports the respective fitted regression coefficients $\hat{\gamma}$ and HAC standard errors in parentheses. The algorithm classifies the country data into six subgroups. The leading five of these subgroups form convergence clubs. In each case the fitted regression coefficient is significantly positive, revealing strong empirical support for the club classification. The final group has a fitted coefficient that is significantly negative, thereby rejecting convergence and revealing evidence of divergence. For clubs 1 through 5, while the point estimates of γ are all significantly positive, they are also significantly less than 2.0. So there is strong evidence of conditional convergence but little evidence of level convergence within each of these clubs.

The middle panel (headed 'Tests of club merging') reports the tests conducted to determine whether any of the original subgroups can be merged to form larger convergence clubs. We

Table II. Convergence club classification: 152 PWT countries from 1970 to 2003

Initial classification $\hat{\gamma}$ (SE of $\hat{\gamma}$)		Tests of club merging $\hat{\gamma}$ (SE of $\hat{\gamma}$)				Final classification $\hat{\gamma}$ (SE of $\hat{\gamma}$)	
Club 1 [50]	0.382 (0.041)	Club 1+2 −0.051*				Club 1 [50]	0.382 (0.041)
Club 2 [30]	0.240 (0.035)	(0.023)	Club 2+3 −0.104*			Club 2 [30]	0.240 (0.035)
Club 3 [21]	0.110 (0.032)		(0.016)	Club 3+4 −0.192*		Club 3 [21]	0.131 (0.064)
Club 4 [24]	0.131 (0.064)			(0.038)	Club 4+5 −0.044*	Club 4 [38]	−0.044 (0.070)
Club 5 [14]	0.190 (0.111)				(0.070)	Club 5 + Group 6 −0.881*	
Group 6 [13]	−1.116* (0.060)					(0.027)	Group 5 [13]	−1.116* (0.060)

Note: * Reject the null hypothesis of growth convergence at the 5% level. The numbers in brackets stand for the number of countries in a group. The initial clustering suggests five sub-convergence groups and one divergent group. Tests for club mergers lead to the final classification of four convergence clubs and one divergence group.

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consider adjacent subgroups in the original classification and each cell in the panel reports the fitted regression coefficient and corresponding HAC standard error. Except for clubs 4 and 5, there is no evidence to support mergers of the original groupings. Hence, the first three subgroups are taken to form separate convergence clubs, while the aggregate of subgroups 4 and 5 constitute a fourth but somewhat weaker convergence club. Note that the point estimate $\hat{\gamma} = -0.044$ of γ is negative but is not statistically significant. In last subgroup (6), there appears to be no evidence of any convergent clubs and no evidence of merging behavior with club 5. The right panel of Table II gives the final empirical classification from this clustering analysis into four growth convergence clubs and one divergent subgroup of countries. Below we report evidence in support of transitioning between these clubs and possible transitioning over time.

Figure 5 displays the country names arranged according to each of the four convergence clubs and the divergent group. The first convergence club (Club 1) consists of a large group of 50 countries. The second convergence club (Club 2) consists of 30 countries. The third convergence club (Club 3) consists of 21 countries. The last convergence club (Club 4) consists of 38 poor countries. Group 5 consists of 13 depressed economies. Given recent historical experience, several nations in Club 4 might be expected to transition to group 5 once data are extended to 2007.

This clustering result and the methodology differ from previous empirical studies on growth convergence clubs. Durlauf and Johnson (1995) used regression tree analysis to produce four locally converging groups of countries. Hobijn and Franses (2000) produced regional converging groups by applying KPSS tests to all pairs of relative log income differences, and Canova (2004) used the predictive density of the data to produce several growth clusters among European regions.

The differences between these studies and the present work primarily arise from our use of the clustering algorithm in Phillips and Sul (2007), which focuses on how idiosyncratic transitions behave over time in relation to the common growth component. To highlight the implications of this transitional approach, Figure 6 plots final period log income in 2003 against initial period log income in 1970. The bold line in the figure is the 45-degree line, so that the distance between each point and this line implies the average growth rate over 34 years. The growth rates of Club 1 are seen to be much higher than those of Club 2 in general, but the final period and initial incomes of many countries in Club 1 are lower than those of Club 2. Similar patterns are observed between Clubs 2 and 3, and between Clubs 3 and 4. Most of the members in the divergent group 5 show evidence of negative economic growth.

As in Durlauf and Johnson (1995) and Canova (2004), one might investigate whether β-convergence occurs within the various convergence clubs by conducting non-augmented Solow regressions of the form

$$\frac{\log y_{iT} - \log y_{i1}}{T-1} = a + \beta \log y_{i1} + \varepsilon_i \tag{20}$$

and performing regression tests on β. Then β-convergence within a cluster will apply if (i) within that club the transition parameter converges (i.e., $b_{it} \to b$, which in turn implies that $x_{it} \to x$), and (ii) the speed of technological learning is faster than the speed of technology creation. On the other hand, β-convergence does not necessarily imply relative convergence of technological progress. For example, suppose that $x_{it} = x_i$ for all t and i, but that $x_i > x_j$ in the long run, whereas initial

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Figure 5. Clustering analysis and convergence clubs: 152 PWT countries from 1970 to 2003. This figure is available in color online at www.interscience.wiley.com/journal/jae

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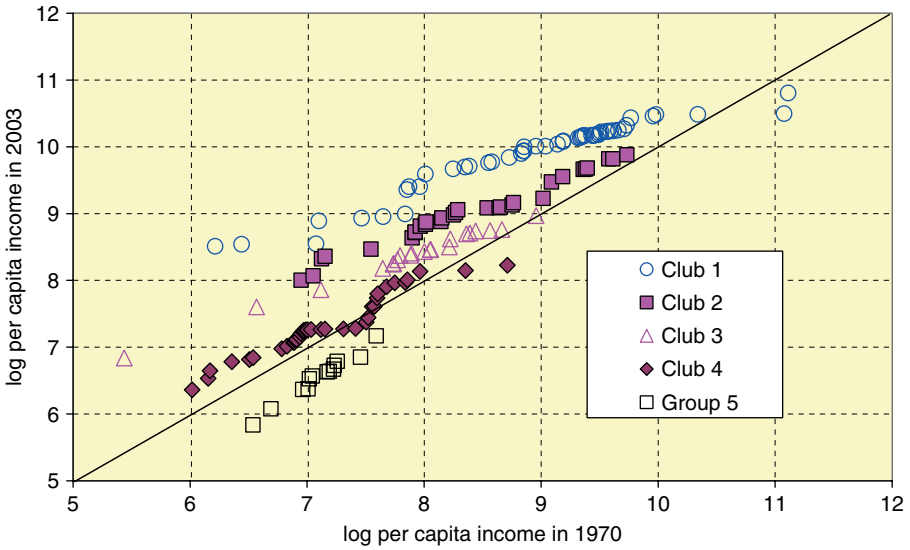


Figure 6. Initial and final period incomes across convergent and divergent groups (152 PWT countries from 1970 to 2003). This figure is available in color online at www.interscience.wiley.com/journal/jae

period income satisfies $\log y_{i1} < \log y_{j1}$. In that case, (20) suggests that β-convergence will occur. Yet the technological differential $x_i > x_j$ leads to divergence in the long run.[16]

Figure 7 plots the average growth rates against per capita real income in 1970 together with point estimates of β. First, we ran a non-augmented Solow regression as in (20) with 152 countries, finding $\hat{\beta} = -0.0004$ with a t-statistic of -0.237. Next, we ran these Solow regressions for each convergence club and the divergence group. For all clubs, $\hat{\beta}$ was found to be significantly less than zero at the 5% level. It is important to note that these negative values of β do not always imply convergence. Among the divergence group, for instance, the point estimate of β was the largest in absolute value, but there is no evidence for convergence in that group.

It is clearly of interest to investigate the general characteristics of these various convergence and divergence subgroups as well as the many possible determining factors in each case, but this analysis would take us well beyond the scope of the current paper.

5.2. Economic Growth Transition

As the world economy and its constitution of nation states change, we may anticipate some evolution in the group clusters over time. The presence of five distinct groups indicates substantial diversity in economic performance among countries and raises the possibility of some transitioning between the groups. Many paths of transition are possible, depending on the individual circumstances of each country, its exposure to common world technology, and the extent to which it may have experienced or be undergoing political upheaval and ethnic or social conflict. The present

[16] A more detailed discussion of the relations between cross-section heterogeneity of initial income, speed of convergence, and growth rate differentials is given in Phillips and Sul (2003).

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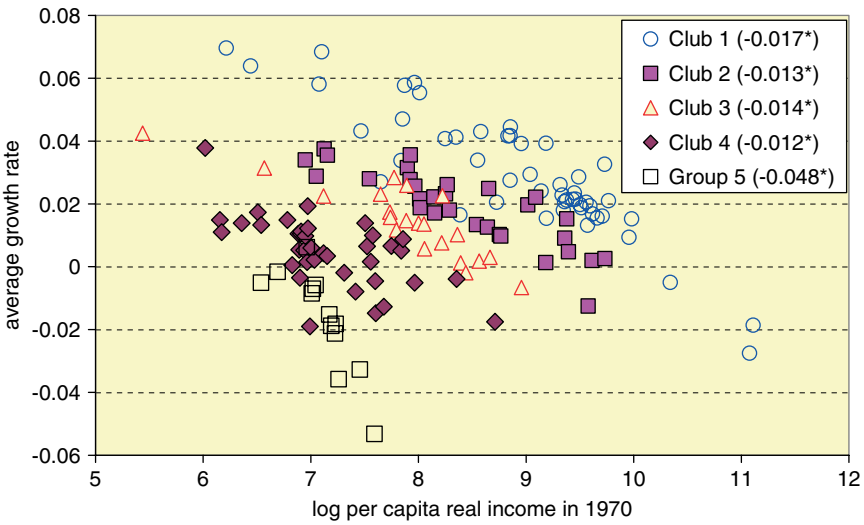


Figure 7. β-Convergence and convergent clubs (152 PWT countries from 1970 to 2003). Numbers in parentheses stand for the estimated regression coefficients β on initial period log income. An asterisk indicates that the point estimates are significant at the 5% level. For all 152 countries, the point estimate of β is -0.0004 and its t-ratio is -0.237. This figure is available in color online at www.interscience.wiley.com/journal/jae

section explores empirical evidence in the world economy for such transitioning in economic performance over time.

The clustering procedure outlined in the previous subsection allows us to examine evidence for transitioning between groups, or the possibility of sequential club convergence where part of one group moves towards another group. In particular, we can use the log t-test to assess evidence of convergence between neighboring members of different club clusters. This test is performed by running the log t-test regression using data that includes a fraction (λ_1) of the lowest (in terms of final income) members in the upper club together with a fraction (λ_2) of the highest (in terms of final income) members in the lower club. In our present application, we set $\lambda_1 = \lambda_2 = 0.5$.

Table III shows the results of this empirical analysis. Panel A reports results of log t convergence tests across groups. The findings indicate strong evidence of transitioning across the top club clusters. In particular, the data show support for conditional convergence between the 15 highest countries in Club 2 and the 25 lowest countries in Club 1 ($\hat{\gamma} = 0.465$, SE ($\hat{\gamma}$) $= 0.049$). Similarly, there is support for conditional convergence between the 15 lowest countries in Club 2 and the 10 highest countries in Club 3 ($\hat{\gamma} = 0.554$, SE ($\hat{\gamma}$) $= 0.055$). Thus, these groups may be understood to be in a state of transition. There is no empirical evidence of transitioning behavior between Club 4 and Club 3. This may be explained by the fact that most of the economies in Club 4 have suffered from various internal and external conflicts and political and social instability, as discussed above.

Some comments on the interpretation of transition behavior are in order. One possible interpretation is that, although certain club clusters may have been identified and overall convergence among countries may have been rejected with given sample data, there remains the possibility that certain clubs may be slowly converging to one another. Another interpretation, and one that may be more realistic in the presence of substantial income diversity across countries, is

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Table III. Economic growth transitions

Panel A. Transition between clubs (152 PWT countries from 1970 to 2003)

Club numbers	$\hat{\gamma}$ (SE of $\hat{\gamma}$)
Club 1 [lower 25] + Club 2 [upper 15]	0.465 (0.049)
Club 2 [lower 15] + Club 3 [upper 10]	0.554 (0.055)
Club 3 [lower 11] + Club 4 [upper 19]	−0.153* (0.019)

Panel B. Club transition over time (98 PWT countries from 1960 to 2003)

From 1960 to 1985: $\hat{\gamma}$ (SE of $\hat{\gamma}$)		From 1970 to 2003: $\hat{\gamma}$ (SE of $\hat{\gamma}$)	
Club 1 [25]	0.343 (0.060)	Club 1 [33]	0.317 (0.057)
Club 2 [10]	0.229 (0.090)	Club 2 [18]	0.228 (0.038)
Club 3 [12]	0.255 (0.072	Club 3 [14]	0.165 (0.034)
Club 4 [19]	0.174 (0.072)	Club 4 [26]	0.045 (0.075)
Group 5 [32]	−0.687 (0.007)*	Club 5 [7]	0.170 (0.055)

Note: * Reject the null hypothesis of growth convergence at the 5% level. The numbers in brackets stand for the number of countries in a group.

that some countries within a certain convergence group may exhibit a tendency to be in transition towards a higher or lower club, leaving open the possibility of joining the new club in the future.

To examine evidence of the latter form of transitioning, we repeat the exercise above by resetting the data initialization from 1970 to 1960 so as to achieve a longer time series sample. In making this reinitialization, one third of the 152 PWT countries drop from the sample, giving a new country count in the subgroups as follows: Club 1 (17), Club 2 (12), Club 3 (7), Club 4 (12), and Group 5 (6). The results are shown at the left of Panel B in Table III. Using this new panel of 98 countries over the period from 1960 to 1985, we find evidence to support the conclusion of four convergence clubs and a fifth divergent group, giving results that broadly correspond to those for the period 1970–2003. In reaching this determination, however, we find that the boundaries between the groups is not as clear for this earlier subperiod. As before, the algorithm originally classifies the country data into six subgroups and then some reduction is achieved on a further pass through the data, leading to four clubs and a residual divergent group. Taking the first two clubs together (Clubs 1 and 2) the regression test does not reject the null of convergence, but the test does reject the inclusion of Club 3 into this larger group. Similar patterns apply with subgroups 2 and 3, subgroups 3 and 4, and subgroups 4 and 5. Thus, there is greater ambiguity about the club boundaries for the period 1960–1985. In making the final determination, we have followed the previous rule and classified subgroups 1, 2, 3 as Clubs 1, 2 and 3, while classifying the sum of subgroups 4 and 5 as Club 4, and leaving the residual group as a divergent subgroup.

The right part of Panel B in Table III reports the results of applying the same clustering algorithm to the same 98 PWT countries over the period 1970–2003. We now find evidence of five clubs overall. Over time, the number count in each of the first four clubs has risen and the number of countries in group 5 has fallen. Moreover, the seven countries in group 5 satisfy the growth convergence test ($\hat{\gamma} = 0.170$, $\mathrm{SE}(\hat{\gamma}) = 0.055$), so that this group is now classified as Club 5.

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Figure 8 and 9 summarize the transitional movements across country groups over time. These figures list the countries in each club based on the clustering results obtained for the period between 1960 and 1985. The figures further display the changes in club membership that have taken place by 2003. Figure 8 focuses on the transitions to Club 1, while Figure 9 tracks the remaining transitions. There are 10 new members in Club 1. About 40% of the countries in Club 2 move to Club 1 over the later time period. Thailand moved from Club 3 to Club 1, and three countries (Cape Verde, Chile and China) jumped to Club 1 from Club 4. Finally, Equatorial Guinea jumped from Group 5 to Club 1. There are 12 new members in Club 2 comprising seven countries from Club 3, four countries from Club 4, and India, which moved from Group 5 to Club 2. Similarly, there are significant changes in the membership of Clubs 3 and 4. One country, the Congo Republic, slipped to Club 4 from Club 3. Meanwhile, seven countries from Club 4 and three countries from Group 5 joined Club 3. Except for the seven countries that remained in Group 5, 21 countries in this group joined Club 4.

In sum, therefore, there is some strong empirical support for economic transitioning and evolving membership of convergence clubs. Only for those particularly poor countries where overall growth rates are negative is there no evidence of economic transition from Group 5 to a convergence club. These countries seem to be caught in a 'phase A' pattern of transitional divergence and seem not to be sharing in the global common growth component. With the short time series sample used here, of course, it is difficult to assess whether or not the negative growth sustained by some

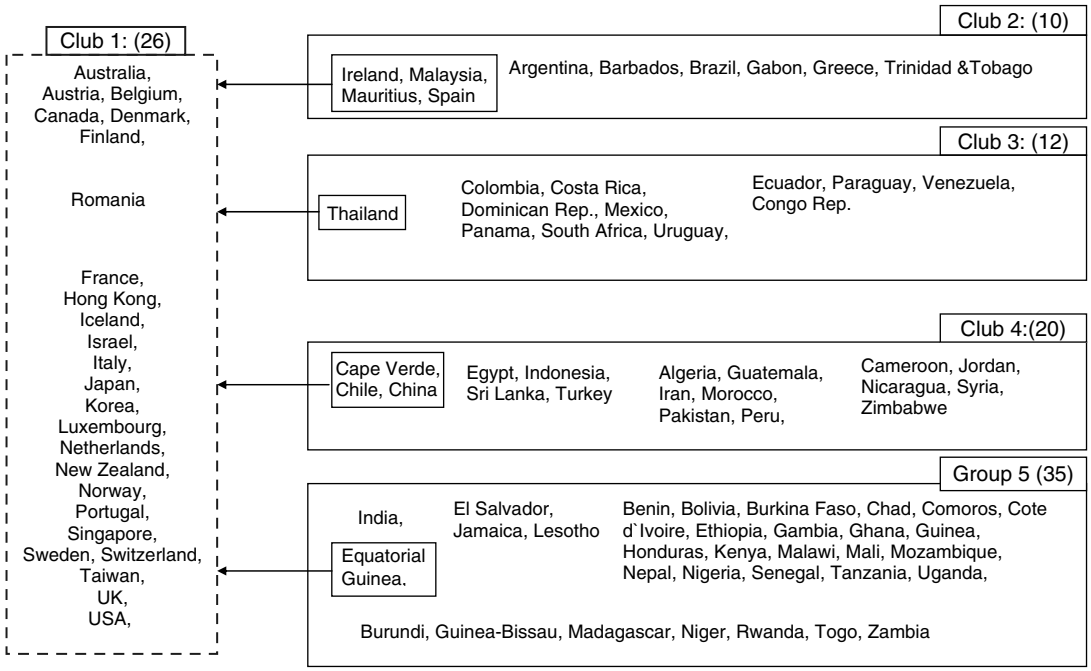


Figure 8. Transitioning to Club 1

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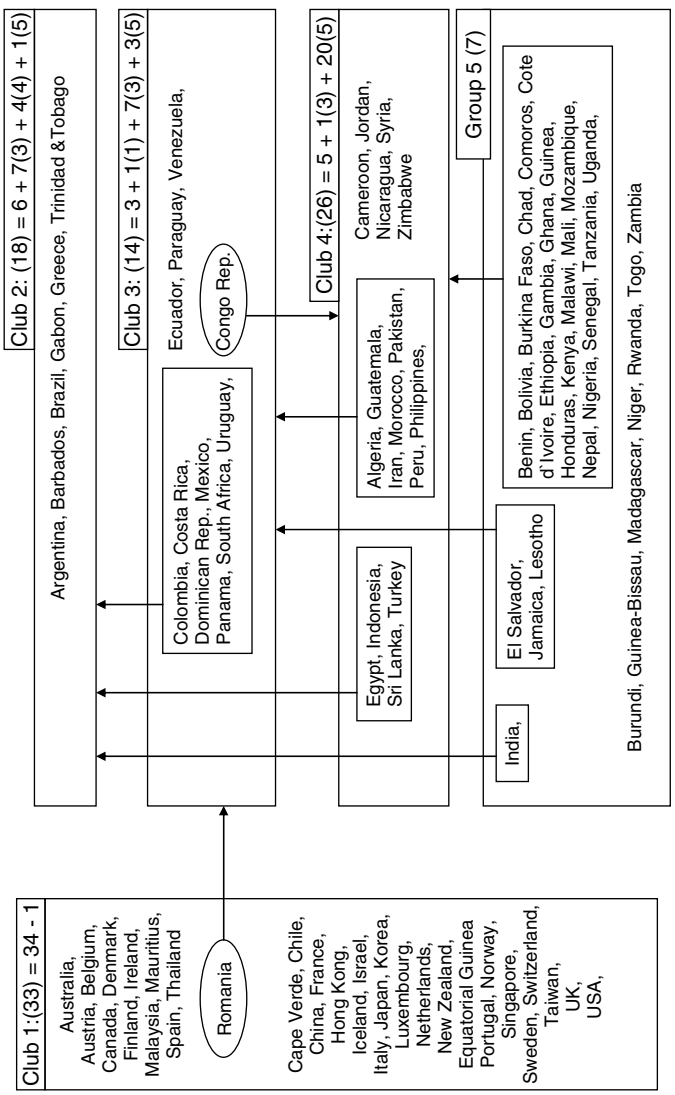


Figure 9. Transitioning to Clubs 2, 3 and 4

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sub-Saharan countries is permanent or transitory. By contrast, there is ample empirical evidence that emerging countries such as the Eastern European countries and parts of the former Soviet Union have shown dramatic *U*-shape transition patterns of real income dynamics during the sample period, undergoing all three phases of transition in this short period.

6. CONCLUSION

As authors such as Durlauf and Quah (1999) have noted, the study of cross-country economic growth often reveals more about heterogeneity in economic performance than it does about convergence. Indeed, just as the distribution of income within nations displays inequality that evolves over time, the distribution of income across nations moves over time, often in ways that cannot be anticipated. Nonetheless, it is also evident that the benefits of modern technology are spreading across national borders and influencing economic performance. of course, this diffusion occurs more quickly in some cases and for some countries than it does for others. Thus, while there are good reasons to expect some convergence in economic performance, especially with the growth of regional economic unions, there are also reasons to expect that the paths of transition in economic performance may be very different across nations. Indeed, in the process of observing nations over time, we observe many different forms of transitional behavior. Some groups of countries or economic regions behave in a similar way over time and appear to be moving on a path towards some steady-state growth pattern. Others appear to diverge over certain periods of time, fall behind and then turn around and show evidence of catching up.

This paper provides some mechanisms for thinking about such transitions, modeling them in a manner that is compatible with a neoclassical framework, and measuring them econometrically. To do so, we focus not on individual economic growth but on economic growth relative to the average performance in a subgroup of economies or an individual benchmark like that of the US economy, which is relevant because it provides for an underlying growth pattern based on the latest common technology. This process enables us to identify the relative transitions that occur within these subgroups and to measure these transitions against the correlative of a common growth trend. Thus, in measuring a country's economic transition curve, we are able to assess its path over time relative to a useful benchmark. In this approach, the transition curve of an economy is an individual characteristic, allowing for many ways in which a neoclassical steady state can be approached, including the possibility of growth convergence clusters or even transitional divergence from the steady state.

The reality of economic transition raises questions about the relevant factors that influence transition. Just as a host of variables have been considered in analyzing the determinants of growth (Barro, 1997; Sala-i-Martin, 1997), there are similarly a large number of factors potentially influencing transition. These factors range over the many economic, social, cultural and political facets that characterize individual countries. In ongoing work the authors are using the methods of this paper to explore the manner in which such factors may influence economic transition. Just as there is transition in economic growth performance over time, we may also expect transition behavior (relative to some benchmark) in many of the factors that influence growth, such as human capital and educational attainment. Linkages between these two forms of transitional behavior contribute to our understanding of the time-forms of long-run economic performance and the various transitions that individual economies experience. The methodology of the present paper is intended to advance this fundamental enquiry.

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7. TECHNICAL APPENDIX

Write the production function in the neoclassical theory of growth with labor augmented technological progress as $Y = F(K, LHA)$ and define

$$\tilde{y} = f(\tilde{k}),\ \tilde{y} = Y/LHA,\ \tilde{k} = K/LHA,\ y = \tilde{y}HA = \tilde{y}A \tag{21}$$

where Y is total output, L is the quantity of labor input, H is the stock of human capital, A is the state of technology, K is physical capital, and $\tilde{y}$ is output per effective labor unit. In the last part of (21), H is normalized to unity so that technology A is defined broadly to encompass the effects of human capital.

It is commonly assumed that technological progress follows a simple exponential path of the form

$$A_{it} = A_{i0}e^{xt} \tag{22}$$

where the growth rate of technology is common across countries. After imposing homogeneity restrictions on $A_i(0)$ and assuming a Cobb–Douglas production function, the transition dynamics of real per capita income have the form

$$\log y_{it} = \log \tilde{y}_i^* + [\log \tilde{y}_{i0} - \log \tilde{y}_i^*]e^{-\beta t} + \log A_0 + xt \tag{23}$$

where $\tilde{y}_i^*$ is the steady-state level of real effective per capita income and β is the speed of convergence given by $(1-\alpha)(n_i + x + \delta)$. Note that α is the capital share, n_i is the population growth rate and δ is the depreciation rate.

Under heterogeneous technological progress with constant population growth and saving rates, the capital stock is determined by

$$\dot{k}_{it} = s_i k_{it}^{\alpha} - (n_i + x + \delta)k_{it} \tag{24}$$

where s_i is the saving rate. The steady-state value k_i^* is obtained by setting $\dot{k}_{it} = 0$, giving $k_i^* = (s_i/[n_i + x + \delta])^{1/(1-\alpha)}$.

The local transition path for $\log k_t$ in the neighborhood of the steady state is obtained by taking the first-order Taylor expansion of (24), or equivalently

$$\frac{\dot{k}_{it}}{k_{it}} = s_i k_{it}^{\alpha-1} - (n_i + x + \delta)$$

about $\log k_i^*$. We have

$$s_i k_{it}^{\alpha-1} - sk^{*\alpha-1} \simeq s_i(\alpha-1)k_i^{*\alpha-1}(\log k_{it} - \log k_i^*) = -\beta(\log k_{it} - \log k_i^*), \text{ say,}$$

with $\beta = (1-\alpha)s_i k_i^{*\alpha-1} = (1-\alpha)(n_i + x + \delta)$. Hence, the local transition path is governed by the linear system

$$\dot{k}_{it}/k_{it} = -\beta(\log k_{it} - \log k_i^*) \tag{25}$$

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When technological progress is heterogeneous, the steady-state value k_i^* does not change but the transition path for $\log k_{it}$ now depends on the transition path for x_{it}

$$\dot{k}_{it}/k_{it} = s_i k_{it}^{\alpha-1} - (n_i + x_{it} + \delta) = s_i k_{it}^{\alpha-1} - (n_i + x + \delta) - (x_{it} - x)$$

As above, we have the expansion

$$s_i k_{it}^{\alpha-1} - (n_i + x + \delta) \simeq -\beta(\log k_{it} - \log k_i^*)$$

which leads to the following dynamic system governing the local transition path about the steady state:

$$\dot{k}_{it}/k_{it} = -\beta(\log k_{it} - \log k_i^*) - (x_{it} - x) \tag{26}$$

For an advanced economy where the technological growth rate has stabilized we have $x_{it} = x$, in which case the transitional path dynamics (26) is the same as (25). For a less developed economy, however, the general solution of (26) gives the following transitional path:

$$\log k_{it} = \log k_i^* + d_{i0} e^{-\beta t} - e^{-\beta t} \int_0^t e^{\beta p}(x_{ip} - x)\mathrm{d}p$$

which depends on the full past trajectory $\{x_{ip}\}_{p \le t}$ of the technology parameter and its deviation from x. For $t = 0$, we have the initialization $\log k_{i0} = \log k_i^* + d_0$, so that $d_{i0} = (\log k_{i0} - \log k_i^*)$, giving the explicit transition path

$$\log k_{it} = \log k_i^* + (\log k_{i0} - \log k_i^*) e^{-\beta t} - e^{-\beta t} \int_0^t e^{\beta p}(x_{ip} - x)\mathrm{d}p \tag{27}$$

Setting $d_{i1} = 1/(\log k_{i0} - \log k_i^*)$, the transitional path (27) has the form

$$\log k_{it} = \log k_i^* + (\log k_{i0} - \log k_i^*) \exp(-\beta_{it} t) \tag{28}$$

where

$$\beta_{it} = \beta - \frac{1}{t} \log \left\{ 1 - d_{i1} \int_0^t e^{\beta p}(x_{ip} - x)\mathrm{d}p \right\} \tag{29}$$

Using this solution, re-express $\log k_{it+q}$ as a function of the initial observation at q, viz., $\log k_{iq}$, so that

$$\log k_{iq} - \log k_i^* = (\log k_{i0} - \log k_i^*) \exp(-\beta_{iq} q)$$

and then

$$\log k_{it+q} = \log k_i^* + (\log k_{iq} - \log k_i^*) \exp(-\beta_{it}^+ t) \tag{30}$$

where $\beta_{it}^+ = \beta_{it+q}(t+q)/t - \beta_{iq} q/t$. Since

$$\beta_{it+q} = \beta - (t+q)^{-1} \log \left\{ 1 - d_{i1} \int_0^{t+q} e^{\beta p}(x_{ip} - x)\mathrm{d}p \right\}$$

$$\beta_{iq} = \beta - q^{-1} \log \left\{ 1 - d_{i1} \int_0^{q} e^{\beta p}(x_{ip} - x)\mathrm{d}p \right\}$$

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we have

$$\beta_{it}^{+} = \beta - t^{-1} \log \left(\left\{ 1 - d_{i1} \int_0^{t+q} e^{\beta p} (x_{ip} - x) \mathrm{d}p \right\} \left\{ 1 - d_{i1} \int_0^{q} e^{\beta p} (x_{ip} - x) \mathrm{d}p \right\}^{-1} \right) \quad (31)$$

In cases where technology is convergent, the deviations $x_{ip} - x$ are bounded and so $\beta_{it}^{+} = \beta + O\left(\frac{\log t}{t}\right) \to \beta$ as $t \to \infty$.

8. DATA APPENDIX

Three panel datasets of log per capita real income are used in the paper. The first panel (A) relates to the 48 contiguous United States from 1929 to 1998 (Source: Bureau of Economic Analysis). The second panel (B) consists of 127 countries from 1950 to 2001. (Source: OECD, *The World Economy: Historical Statistics*). From the same OECD data source we also collected the long historical dataset (the Maddison Dataset) for 18 Western OECD countries covering the period from 1500 to 2001. The third panel (C) includes 152 countries from 1970 to 2003 and 98 countries from 1960 to 2003 (Source: PWT 6.2). The remaining three subsections of this appendix show how subgroups for these panels were created based on regional location and geographical distance.

8.1. Panel Dataset A: Regional US State Classifications

Mid-Altantic: Delaware, Maryland, New Jersey, New York, Pennsylvania
New England: Connecticut, Maine, Massachusetts, New Hampshire, Rhode Island, Vermont
Great Lakes: Illinois, Indiana, Michigan, Ohio, Wisconsin
Mountain: Arizona, Colorado, Idaho, Montana, Nevada, New Mexico, Utah, Wyoming
Pacific: California, Oregon, Washington
Plain States: Iowa, Kansas, Minnesota, Missouri, Nebraska, North Dakota, South Dakota
South Atlantic: Florida, Georgia, North Carolina, South Carolina, Virginia, West Virginia
West South Central: Arkansas, Louisiana, Oklahoma, Texas
East South Central: Alabama, Kentucky, Mississippi, Tennessee

8.2. Panel Dataset B: OECD The World Economy: Historical Statistics

18 Western OECD Countries: Austria, Belgium, Denmark, Finland, France, Germany, Italy, Netherlands, Norway, Sweden, Switzerland, United Kingdom, Portugal, Spain, Australia, New Zealand, Canada, United States

8.3. Panel Dataset C: PWT Version 6.2 Countries

The following subgroups are formed based on geographical location:

Latin American and Caribbean Countries: Argentina, Barbados, Brazil, Chile, Colombia, Costa Rica, Dominican Republic, Ecuador, El Salvador, Jamaica, Mexico, Panama, Paraguay, Trinidad and Tobago, Uruguay, Venezuela, Bolivia, Guatemala, Honduras, Nicaragua, Peru

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Middle East and North African Countries: Algeria, Egypt, Israel, Iran, Morocco, Jordan, Syria
Sub-Saharan Africa Countries: Benin, Burkina Faso, Burundi, Cape Verde, Chad, Comoros, Rep. Congo, Cote d'Ivoire, Equatorial Guinea, Ethiopia, Gabon, Gambia, Ghana, Guinea, Guinea-Bissau, Kenya, Lesotho, Madagascar, Malawi, Mali, Mozambique, Niger, Nigeria, Rwanda, Senegal, South Africa, Tanzania, Togo, Uganda, Zambia, Zimbabwe
19 OECD Countries Australia, Austria, Belgium, Canada, Denmark, Finland, France, Iceland, Italy, Japan, Netherlands, New Zealand, Norway, Portugal, Spain, Sweden, Switzerland, United Kingdom, USA
4 Dragons: Hong Kong, South Korea, Singapore, Taiwan
3 NICs: Indonesia, Malaysia, Thailand

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